

[GETNET]Upgrade Oracle Database 12.1.0.2 to 19c – AIX (StandAlone)

O que nos levou a escrever este artigo são as dúvidas quanto a integridade dos backups do BI, visto que o host GNCASSPL01262 costuma travar e sofrer restart com frequência. Como seu backup demora por volta de 30hs, cada vez mais é recorrente backups INCO quebrados.

Sendo assim, comparamos as queries de banco com os logs do networker backup.

Dicas do Andre Spader, Ioannis Karatziovalis e Igor Carvalho.

1- Validando um backup íntegro no Banco original

Nossa primeira tentativa de restore, ainda não existia o backup do dia 08/06. Em pleno 07/06, o único backup válido era do dia 18/05. Como a primeira tentativa não deu certo, ao tentarmos novamente, já estava disponível o backup do dia 08/06, finalizado em 10/06 e seguimos com ele.

```
SQL> set pages 888
set linesize 500
set serveroutput on
col STATUS for A15
select vw.*
from ( select INPUT_TYPE,SESSION_KEY, SESSION_STAMP, to_char(START_TIME,'dd/mm/yyyy hh24:mi:ss') starttime, to_char(end_time,'dd/mm/yyyy hh24:mi:ss') endtime,
STATUS, round(ELAPSED_SECONDS/60/60) elapsed_hours
from v$RMAN_BACKUP_JOB_DETAILS
order by START_TIME desc) vw
where INPUT_TYPE = 'DB INCR';
```

INPUT_TYPE	SESSION_KEY	SESSION_STAMP	STARTTIME	ENDTIME	STATUS	ELAPSED_HOURS
DB INCR	1068546	1138981514	08/06/2023 15:46:04	10/06/2023 02:51:25	COMPLETED	35
DB INCR	1067715	1138376720	01/06/2023 15:46:44	FAILED		
DB INCR	1066933	1137771917	25/05/2023 15:46:45	FAILED		
DB INCR	1066094	1137167121	18/05/2023 15:46:22	19/05/2023 22:45:14	COMPLETED	31

De posse desta data inicio e data fim, cruzei as informações com os logs gerados nos servidores:

```
[oracle@gncasspl01262 logs]$ pwd
/u02/scripts/backup/rman/logs

[oracle@gncasspl01262 logs]$ ls -ltr *INC*
...
-rw-r--r-- 1 oracle oinstall 1324086 Jun 10 03:12 RMAN_BI_INCO_20230608_1545.log
NOTA: dentre vários, aquele com data parecida
```

Note que em todos os arquivos de log, este, em seu nome, se inicia no horário próximo ao esperado da query do banco e finaliza (horário do arquivo) próximo ao fim do backup.

2 – Analisando o Log gerado

Logo no começo do log, podemos pegar o DB_ID do banco:

```
connected to target database: BI (DBID=933493156)
```

Uma dica valiosa do Ioannis é buscar por "including" no vi, assim podemos pegar os TAGs dos datafiles e TAG do controlfile:

```
piece handle=BI_933493156_652058_1139099988_20230610 tag=TAG20230608T154624 comment=API Version 2.0,MMS Version 19.7.0.0
channel t3: backup set complete, elapsed time: 01:17:36
channel t4: finished piece 1 at 10-JUN-23
piece handle=BI_933493156_652057_1139099303_20230610 tag=TAG20230608T154624 comment=API Version 2.0,MMS Version 19.7.0.0
channel t4: backup set complete, elapsed time: 01:40:03
channel t5: finished piece 1 at 10-JUN-23
piece handle=BI_933493156_652066_1139103790_20230610 tag=TAG20230608T154624 comment=API Version 2.0,MMS Version 19.7.0.0
channel t5: backup set complete, elapsed time: 00:56:26
channel t2: finished piece 1 at 10-JUN-23
piece handle=BI_933493156_652065_1139103553_20230610 tag=TAG20230608T154624 comment=API Version 2.0,MMS Version 19.7.0.0
channel t2: backup set complete, elapsed time: 01:00:42
channel t1: finished piece 1 at 10-JUN-23
piece handle=BI_933493156_652064_1139102277_20230610 tag=TAG20230608T154624 comment=API Version 2.0,MMS Version 19.7.0.0
channel t1: backup set complete, elapsed time: 01:32:38
Finished backup at 10-JUN-23
```

```
Starting backup at 10-JUN-23
channel t1: starting full datafile backup set
channel t1: specifying datafile(s) in backup set
including current control file in backup set
channel t1: starting piece 1 at 10-JUN-23
channel t1: finished piece 1 at 10-JUN-23
piece handle=p81uaq02_652072_1_1 tag=TAG20230610T025042 comment=API Version 2.0,MMS Version 19.7.0.0
channel t1: backup set complete, elapsed time: 00:00:15
Finished backup at 10-JUN-23
```

```
Starting Control File and SPFILE Autobackup at 10-JUN-23
piece handle=c-933493156-20230610-02 comment=API Version 2.0,MMS Version 19.7.0.0
Finished Control File and SPFILE Autobackup at 10-JUN-23
```

Agora temos certeza, ao ler o log inteiro, que o backup foi feito com sucesso neste dia, assim como indicado o banco e confirmado no arquivo de log.

3- Restore:

Todas as ações são no host novo, não confundir com a origem!

Criando o db_profile:

```
[oracle@gncasspl08329 ~]$ cat db_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1
export ORACLE_SID=BI
export OH=$ORACLE_HOME
export JAVA_HOME=/usr/bin/java
export GGS_HOME=/ggs
export PATH=.:/usr/bin/java/bin:/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/oracle/bin:$ORACLE_HOME/bin:$ORACLE_HOME/OPatch:$PATH
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib:$ORACLE_HOME/network/jlib
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$ORACLE_HOME/oracm/lib:/lib:/usr/lib:/usr/local/lib
export LIBPATH=/ggs:$ORACLE_HOME/lib:/usr/lib:$LIBPATH
```

Criando um pfile baseado no BI produtivo, GNCASSPL01262:

```

bash-4.3$ cat /u01/app/oracle/product/12.1.0/dbhome_1/dbs/initBI.ora
*._data_transfer_cache_size=0
*._db_cache_size=187904819200
*._java_pool_size=3758096384
*._large_pool_size=3758096384
*._oracle_base='/u01/app/oracle'#ORACLE_BASE set from environment
*._pga_aggregate_target=171798691840
*._sga_target=214748364800
*._shared_io_pool_size=536870912
*._shared_pool_size=15569256448
*._streams_pool_size=2684354560
*.audit_file_dest='/u01/app/oracle/product/admin/BI/adump'
*._allow_resetlogs_corruption=TRUE
*.audit_trail='NONE'
*.compatible='12.1.0.2.0'
*.control_files='+DATA/bidcs/controlfile/current.788.972588957','+ARCH/bidcs/controlfile/current.722.972588957','+DATA/bidcs/controlfile/current.744.972588957'#Restore Controlfile
*.db_block_size=8192
*.db_domain=""
*.db_files=6000
*.db_name='BI'
*.db_recovery_file_dest=""
*.db_recovery_file_dest_size=1
*.diagnostic_dest='/u01/app/oracle/product'
*.dispatchers='(PROTOCOL=TCP) (SERVICE=BIXDB)'
*.open_cursors=1500
*.pga_aggregate_target=171798691840
*.processes=1500
*.sessions=1655
*.undo_tablespace='UNDOTBS2'
*.remote_login_passwordfile='EXCLUSIVE'
*.sga_target=214748364800

```

No ambiente novo, os diskgroups foram criados com os mesmos nomes do host original:

```

SQL> set lines 300
SELECT name, type, total_mb, free_mb, required_mirror_free_mb,
usable_file_mb FROM V$ASM_DISKGROUP;

```

NAME	TYPE	TOTAL_MB	FREE_MB	REQUIRED_MIRROR_FREE_MB	USABLE_FILE_MB
DATA_AUX	EXTERN	51200	51052	0	51052
ARCH	EXTERN	5242860	5241488	0	5241488
DATA	EXTERN	136314360	136310392	0	136310392
GRID	EXTERN	15356	15260	0	15260

Criando o apontamento no tnsnames para o catálogo do RMAN:

```

[oracle@gncasspl08329 admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/12.1.0.2/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

LISTENER_BI =
  (ADDRESS = (PROTOCOL = TCP)(HOST = gncashpl05340.getnet.local)(PORT = 1521))

BI =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = gncashpl05340.getnet.local)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = BIQA)
    )
  )

##### Novo Backup Networker
RCAT=(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=vip-wht-rmancat18c-db-oracle.getnet.local)(PORT=1521))(CONNECT_DATA=(S
ERVER=DEDICATED)(SERVICE_NAME=rcat18chDCN)))

```

Criando diretórios que são citados no pfile:

```
[oracle@gncasspl08329 dbs]$ mkdir -p /u01/app/oracle/product/admin/BI/adump
[oracle@gncasspl08329 dbs]$ mkdir -p /u01/app/oracle
[oracle@gncasspl08329 dbs]$ mkdir -p /u01/app/oracle/product
```

Subindo o banco em nomount:

```
[oracle@gncasspl08329 dbs]$ sqlplus / as sysdba
```

SQL*Plus: Release 12.1.0.2.0 Production on Wed Jun 7 17:05:56 2023

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Connected to an idle instance.

```
SQL> startup nomount pfile='/u01/app/oracle/product/12.1.0.2/dbhome_1/dbs/initBI.ora';
ORACLE instance started.
```

Total System Global Area 2.1475E+11 bytes

Fixed Size 7658152 bytes

Variable Size 3.2749E+10 bytes

Database Buffers 1.8146E+11 bytes

Redo Buffers 529211392 bytes

SQL>

Exportando variáveis do RMNA.

NOTA: tirei as variáveis do arquivo de configuração padrão fornecido pela equipe de backup ao configurar o backup no host.

```
[oracle@gncasspl08329 ~]$ cd /u02/scripts/backup/rman/
[oracle@gncasspl08329 rman]$ ls -la
total 36
drwxrwx--- 3 oracle oinstall 4096 Jun 6 16:25 .
drwxrwx--- 3 oracle oinstall 4096 Jun 6 16:25 ..
-rwxr-xr-x 1 oracle oinstall 333 Jun 6 16:25 BIQA_ARCH_nmda.cfg
-rwxr-xr-x 1 oracle oinstall 100 Jun 6 16:25 BIQA_GNCASSPL08329_ARCH_DCN_PROD.rman
-rwxr-xr-x 1 oracle oinstall 100 Jun 6 16:25 BIQA_GNCASSPL08329_INC0_DCN_PROD.rman
-rwxr-xr-x 1 oracle oinstall 101 Jun 6 16:25 BIQA_GNCASSPL08329_MAINT_DCN_PROD.rman
-rwxr-xr-x 1 oracle oinstall 333 Jun 6 16:25 BIQA_INC0_nmda.cfg
-rwxr-xr-x 1 oracle oinstall 334 Jun 6 16:25 BIQA_MAINT_nmda.cfg
drwxrwx--- 2 oracle oinstall 4096 Jun 6 16:24 logs
```

```
[oracle@gncasspl08329 rman]$ cat BIQA_ARCH_nmda.cfg
ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1
ORACLE_SID=BIQA
ORACLE_USER=oracle
NLS_LANG=AMERICAN_AMERICA.WE8MSWIN1252
TNS_ADMIN=ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1/network/admin/
NSR_RMAN_ARGUMENTS=msglog '/u02/scripts/backup/rman/logs/RMAN_BIQA_ARCH_`date +%Y%m%d_%H%M`.log' append
NSR_RMAN_OUTPUT=SHELL
```

Export em si:

```
export ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1
export ORACLE_SID=BI
export ORACLE_USER=oracle
export NLS_LANG=AMERICAN_AMERICA.WE8MSWIN1252
export TNS_ADMIN=ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1/network/admin/
export NSR_RMAN_ARGUMENTS=msglog '/u02/scripts/backup/rman/logs/RMAN_BIQA_ARCH_`date +%Y%m%d_%H%M`.log' append
export NSR_RMAN_OUTPUT=SHELL
```

3.1 – Restore do Control File

O primeiro passo do restore do banco é restaurar os arquivos do Control File. Conforme explicado anteriormente, o Control File possui um TAG específico que já foi identificado.

Configuração do script de restore do Control File:

NOTA: DBID pegamos no log, TAG pegamos no log, NSR_CLIENT, é o host original:

```
[oracle@gncasspl08329 ~]$ rman

Recovery Manager: Release 12.1.0.2.0 - Production on Thu Jun 8 09:39:39 2023

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RMAN> connect target /

connected to target database: BI (not mounted)

RMAN> connect catalog rmancat/rmancat@RCAT

connected to recovery catalog database
recovery catalog schema release 19.12.00.00. is newer than RMAN release

RMAN> SET DBID 933493156;

executing command: SET DBID
database name is "BI" and DBID is 933493156

RMAN> run
{
  ALLOCATE CHANNEL t1 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncaswn05977.wt.getnet.local,NSR_CLIENT=gncasspl01
262.wt.getnet.local)';
  restore controlfile from tag 'TAG20230610T025042';
  alter database mount;
  release channel t1;
}

allocated channel: t1
channel t1: SID=793 device type=SBT_TAPE
channel t1: NMDA Oracle v19.4.0.3

Starting restore at 10-JUN-23

channel t1: starting datafile backup set restore
channel t1: restoring control file
channel t1: reading from backup piece p81uaq02_652072_1_1
channel t1: piece handle=p81uaq02_652072_1_1 tag=TAG20230610T025042
channel t1: restored backup piece 1
channel t1: restore complete, elapsed time: 00:00:25
output file name=/u01/app/oracle/product/12.1.0.2/dbhome_1/dbs/cntrlBI.dbf
Finished restore at 10-JUN-23

Statement processed

released channel: t1
```

Apenas validando que o Control File foi restaurado no ASM:

```
ASMCMD> pwd
+DATA/BI/CONTROLFILE
ASMCMD> ls
current.266.1138902495
current.267.1138902497
ASMCMD> lsdg
```

State	Type	Rebal	Sector	Logical_Sector	Block	AU	Total_MB	Free_MB	Req_mir_free_MB	Usable_file_MB	Offline_disks	Voting_files
MOUNTED	EXTERN	N	512	512	4096	4194304	5242860	5241296	0	5241296	0	N ARCH/
MOUNTED	EXTERN	N	512	512	4096	4194304	136314360	136310008	0	136310008	0	N DATA/
MOUNTED	EXTERN	N	512	512	4096	4194304	51200	51052	0	51052	0	N DATA_AUX/
MOUNTED	EXTERN	N	512	512	4096	4194304	15356	15260	0	15260	0	N GRID/

3.2 – Restore do banco

Script de restore:

```

#RMAN
source /home/oracle/db_profile
export DATE=`date +%d%m%y`

$ORACLE_HOME/bin/rman target / catalog rmancat@RCAT trace=/home/oracle/restore_BI/rman_refresh_database_from_tape_${DATE}.tr
c log=/home/oracle/restore_BI/rman_refresh_database_from_tape_${DATE}.log << EOF

run{
  ALLOCATE CHANNEL t1 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t2 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t3 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t4 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t5 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t6 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t7 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  ALLOCATE CHANNEL t8 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  SET NEWNAME FOR DATABASE TO '+DATA';
  restore database from tag 'TAG20230518T154635';
  switch datafile all;
  recover database until time "to_date('19/05/2023 23:00:00', 'YYYY-MM-DD HH24:MI:SS')";
  release channel t1;
  release channel t2;
  release channel t3;
  release channel t4;
  release channel t5;
  release channel t6;
  release channel t7;
  release channel t8;
}
EOF

date >> /home/oracle/restore_ecom/rman_refresh_database_from_tape_icmc_${DATE}.log

```

Disparando o script de restore:

```

nohup /home/oracle/restore_BI/rman_refresh_database_from_tape.sh 1>/home/oracle/restore_BI/rman_refresh_database_from_tape.log 2>/home/
oracle/restore_BI/rman_refresh_database_from_tape.err &

```

Uma medida de segurança a mais, é restaurar os Archives do período de duração do backup, para isso, uma query que localiza os arquivos gerados nos dias:

NOTA: os hosts do BI possuem áreas grandes de Archives, não vi problemas em restaurar 3 dias de Archives e deixar por precaução. Porém, não é um passo obrigatório, caso dê erro de falta de Archives, basta restaurar os faltantes.

```
SQL> set pages 1000
SELECT A.*,
Round(A.Count#*B.AVG#/1024/1024) Daily_Avg_Mb
FROM
( SELECT
To_Char(First_Time,'YYYY-MM-DD') DAY, Count(1) Count#, Min(RECID) Min#, Max(RECID) Max#
FROM v$log_history
GROUP BY To_Char(First_Time,'YYYY-MM-DD')
ORDER BY 1 DESC
) A,
( SELECT Avg(BYTES) AVG#, Count(1) Count#, Max(BYTES) Max_Bytes, Min(BYTES) Min_Bytes
FROM v$log
) B;
```

3605182

DAY	COUNT#	MIN#	MAX#	DAILY_AVG_MB
2023-06-12	1785	3606763	3608547	3655680
2023-06-11	1022	3605741	3606762	2093056
2023-06-10	1076	3604665	3605740	2203648 -- fim backup
2023-06-09	2134	3602531	3604664	4370432
2023-06-08	933	3601598	3602530	1910784 -- inicio backup
2023-06-07	2114	3599484	3601597	4329472
2023-06-06	2112	3597372	3599483	4325376
2023-06-05	2263	3595109	3597371	4634624
2023-06-04	1151	3593958	3595108	2357248
2023-06-03	1125	3592833	3593957	2304000
2023-06-02	2586	3590247	3592832	5296128
2023-06-01	2130	3588117	3590246	4362240
2023-05-31	2126	3585991	3588116	4354048
2023-05-30	2245	3583746	3585990	4597760
2023-05-29	2158	3581588	3583745	4419584
2023-05-28	854	3580734	3581587	1748992
2023-05-27	1058	3579676	3580733	2166784
2023-05-26	2250	3577426	3579675	4608000
2023-05-25	2263	3575163	3577425	4634624
2023-05-24	2026	3573137	3575162	4149248
2023-05-23	1925	3571212	3573136	3942400
2023-05-22	1902	3569310	3571211	3895296
2023-05-21	737	3568573	3569309	1509376
2023-05-20	1069	3567504	3568572	2189312
2023-05-19	1971	3565533	3567503	4036608
2023-05-18	2025	3563508	3565532	4147200
2023-05-17	2053	3561455	3563507	4204544
2023-05-16	2249	3559206	3561454	4605952
2023-05-15	2044	3557162	3559205	4186112
2023-05-14	916	3556246	3557161	1875968
2023-05-13	1047	3555199	3556245	2144256
2023-05-12	2015	3553184	3555198	4126720
2023-05-11	2141	3551043	3553183	4384768
2023-05-10	2110	3548933	3551042	4321280
2023-05-09	2125	3546808	3548932	4352000
2023-05-08	2039	3544769	3546807	4175872
2023-05-07	1068	3543701	3544768	2187264
2023-05-06	1120	3542581	3543700	2293760
2023-05-05	2200	3540381	3542580	4505600
2023-05-04	2137	3538244	3540380	4376576
2023-05-03	4948	3533296	3538243	10133504
2023-05-02	3121	3530175	3533295	6391808

40 rows selected.

Restore propriamente dito dos Archives:

NOTA: Rodei referente aos 3 dias envolvidos no backup

NOTA: Rodei referente aos 3 dias envolvidos no backup

2023-06-10	1076	3604665	3605740	2203648 -- fim backup
2023-06-09	2134	3602531	3604664	4370432
2023-06-08	933	3601598	3602530	1910784 -- inicio backup

Script:

```
[oracle@gncasspl08329 ~]$ export ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1
[oracle@gncasspl08329 ~]$ export ORACLE_SID=BI
[oracle@gncasspl08329 ~]$ export ORACLE_USER=oracle
[oracle@gncasspl08329 ~]$ export NLS_LANG=AMERICAN_AMERICA.WE8MSWIN1252
[oracle@gncasspl08329 ~]$ export TNS_ADMIN=ORACLE_HOME=/u01/app/oracle/product/12.1.0.2/dbhome_1/network/admin/
[oracle@gncasspl08329 ~]$ export NSR_RMAN_ARGUMENTS=msglog '/u02/scripts/backup/rman/logs/RMAN_BIQA_ARCH_'date +%Y%m%d_%H%M'.log' append
[oracle@gncasspl08329 ~]$ export NSR_RMAN_OUTPUT=SHELL

[oracle@gncasspl08329 ~]$ rman target /

Recovery Manager: Release 12.1.0.2.0 - Production on Mon Jun 12 09:56:44 2023

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connected to target database: BI (DBID=933493156, not open)

RMAN> connect catalog rmancat/rmancat@RCAT

connected to recovery catalog database
recovery catalog schema release 19.12.00.00. is newer than RMAN release

RMAN> RUN
{
  ALLOCATE CHANNEL t1 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  set archivelog destination to '+ARCH';
  restore archivelog from sequence=3601598 until sequence=3602530;
  release channel t1;
}

RUN
{
  ALLOCATE CHANNEL t1 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  set archivelog destination to '+ARCH';
  restore archivelog from sequence=3602531 until sequence=3604664;
  release channel t1;
}

RUN
{
  ALLOCATE CHANNEL t1 TYPE 'SBT_TAPE' PARMS 'SBT_PARMS=(NSR_SERVER=gncasnwn05977.wt.getnet.local,NSR_CLIENT=gncasspl0
1262.wt.getnet.local)';
  set archivelog destination to '+ARCH';
  restore archivelog from sequence=3604665 until sequence=3605740;
  release channel t1;
}
```

Abrindo o Banco:

```
SQL> alter database open resetlogs;
alter database open resetlogs
*
ERROR at line 1:
ORA-19527: physical standby redo log must be renamed
ORA-00312: online log 100 thread 1:
'+DATA/BIDCS/ONLINELOG/group_100.1801.1003864385'
```

Neste ponto, pelo alert log, vi que precisaria referenciar o db_name, que mudei de BIDCS para BI, sendo assim, baixei o banco e adicionei um parâmetro no init.ora:

```
log_file_name_convert='BIDCS','BI'
```

Ao tentar o abrir o banco, novamente, tomei outro erro, para minha surpresa, o controlfile vê como um banco StandBy, mesmo o restore vindo de um Primary:
NOTA: Ioannis já pegou isso no PNR, segundo ele, poderíamos apenas ativar o Standby.


```
SQL> startup mount
ORACLE instance started.

Total System Global Area 2.1475E+11 bytes
Fixed Size          7658152 bytes
Variable Size       3.2749E+10 bytes
Database Buffers    1.8146E+11 bytes
Redo Buffers        529211392 bytes
Database mounted.
SQL> alter database open resetlogs;
alter database open resetlogs
*
ERROR at line 1:
ORA-01666: control file is for a standby database
```

Seguindo o troubleshooting, segui os passos para ativar um standby, sendo assim, acabei aplicando todos os archives:

```
SQL> select name,open_mode ,database_role from v$database;

NAME      OPEN_MODE      DATABASE_ROLE
-----
BI        MOUNTED        PHYSICAL STANDBY

SQL> alter system set wallet open identified by "xxxxx";

System altered.

SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;

Database altered.

SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

Database altered.

SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE FINISH;

Database altered.

NOTA: este é o comando que aplicou todos os archives que eu tinha restaurado, demorou bastante, devido ao volume.

SQL> ALTER DATABASE ACTIVATE STANDBY DATABASE;

Database altered.

SQL> select name,open_mode ,database_role from v$database;

NAME      OPEN_MODE      DATABASE_ROLE
-----
BI        MOUNTED        PRIMARY

SQL> alter database open;

Database altered.

SQL> select name,open_mode ,database_role from v$database;

NAME      OPEN_MODE      DATABASE_ROLE
-----
BI        READ WRITE     PRIMARY
```

Evidências do banco:

```
SQL> col HOST_NAME format a30
set lines 230
col COMP_NAME format a55
set pages 900
col DESCRIPTION format a70
col ACTION_TIME format a29
col COMMENTS format a40
```

col COMMENT format a9

col ACTION format a9

col NAMESPACE format a9

col ID format 99999999

SQL> select instance_name, host_name, version, status, open_mode, database_role, switchover_status from gv\$instance, v\$database;

INSTANCE_NAME	HOST_NAME	VERSION	STATUS	OPEN_MODE	DATABASE_ROLE	SWITCHOVER_STAT
US						

BI	gncasspl08329.getnet.local	12.1.0.2.0	OPEN	READ WRITE	PRIMARY	NOT ALLOWED
----	----------------------------	------------	------	------------	---------	-------------

SQL> select comp_name, version, status, modified from dba_registry;

COMP_NAME	VERSION	STATUS	MODIFIED
Oracle Application Express	4.2.5.00.08	VALID	18-JUL-2022 20:17:53
OWB	11.2.0.1.0	VALID	18-JUL-2022 20:17:53
OLAP Catalog	11.2.0.4.0	OPTION OFF	08-SEP-2018 10:34:12
Spatial	12.1.0.2.0	VALID	18-JUL-2022 20:17:51
Oracle Multimedia	12.1.0.2.0	VALID	18-JUL-2022 20:17:46
Oracle XML Database	12.1.0.2.0	VALID	18-JUL-2022 20:17:44
Oracle Text	12.1.0.2.0	VALID	18-JUL-2022 20:17:37
Oracle Workspace Manager	12.1.0.2.0	VALID	18-JUL-2022 20:17:13
Oracle Database Catalog Views	12.1.0.2.0	VALID	18-JUL-2022 20:15:55
Oracle Database Packages and Types	12.1.0.2.0	VALID	18-JUL-2022 20:17:06
JServer JAVA Virtual Machine	12.1.0.2.0	VALID	18-JUL-2022 20:17:35
Oracle XDK	12.1.0.2.0	VALID	18-JUL-2022 20:17:36
Oracle Database Java Packages	12.1.0.2.0	VALID	18-JUL-2022 20:17:37
OLAP Analytic Workspace	12.1.0.2.0	VALID	18-JUL-2022 20:17:47
Oracle OLAP API	12.1.0.2.0	VALID	23-JUL-2022 14:40:55

15 rows selected.

SQL> select PATCH_ID, ACTION, STATUS, ACTION_TIME, DESCRIPTION from dba_registry_sqlpatch;

PATCH_ID	ACTION	STATUS	ACTION_TIME	DESCRIPTION
27547329	APPLY	SUCCESS	09-SEP-18 09.31.30.075366 AM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
27547329	ROLLBACK	SUCCESS	02-APR-19 09.43.47.779103 PM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
28731800	APPLY	SUCCESS	02-APR-19 09.43.49.135431 PM	DATABASE BUNDLE PATCH 12.1.0.2.190115
27547329	ROLLBACK	SUCCESS	09-JUN-19 12.54.47.724078 PM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
29141038	APPLY	SUCCESS	09-JUN-19 12.54.49.107027 PM	DATABASE BUNDLE PATCH 12.1.0.2.190416
27547329	ROLLBACK	SUCCESS	08-SEP-19 02.48.39.423453 PM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
29496791	APPLY	SUCCESS	08-SEP-19 02.48.40.866487 PM	DATABASE BUNDLE PATCH 12.1.0.2.190716
27547329	ROLLBACK	SUCCESS	08-DEC-19 01.07.55.296168 PM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
29972716	APPLY	SUCCESS	08-DEC-19 01.07.56.714547 PM	DATABASE BUNDLE PATCH 12.1.0.2.191015
27547329	ROLLBACK	SUCCESS	23-JUL-22 02.41.18.825877 PM	DATABASE PATCH SET UPDATE 12.1.0.2.180717
33711072	APPLY	SUCCESS	23-JUL-22 02.41.20.719475 PM	DATABASE BUNDLE PATCH 12.1.0.2.220419

11 rows selected.

SQL> SELECT TO_CHAR(CURRENT_SCN) FROM V\$DATABASE;

TO_CHAR(CURRENT_SCN)

355227454740

SQL> Select scn_to_timestamp(355227454740) from dual;

SCN_TO_TIMESTAMP(355227454740)

12-JUN-23 10.31.26.000000000 PM

SQL>

Fiz um último Stop/Start para subir o banco com instance name de BIQA. Cadastrei o banco no srvctl, criei um service, spfile e ajuste de monitoria:

```
SQL> shutdown immediate;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> exit
```

Ajuste do init.ora:

```
[oracle@gncasspl08329 dbs]$ pwd
/u01/app/oracle/product/12.1.0.2/dbhome_1/dbs
[oracle@gncasspl08329 dbs]$ mv initBI.ora initBIQA.ora
```

Ajuste do tnsnames.ora:

```
[oracle@gncasspl08329 dbs]$ cd $ORACLE_HOME/network/admin

[oracle@gncasspl08329 admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File: /u01/app/oracle/product/12.1.0.2/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

##### Novo Backup Networker

RCAT =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = vip-wht-rmancat18c-db-oracle.getnet.local)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = rcat18chDCN)
    )
  )

BIQA =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = gncasspl08329.getnet.local)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = BIQA)
    )
  )

LISTENER_BI =
  (ADDRESS = (PROTOCOL = TCP)(HOST = gncashpl05340.getnet.local)(PORT = 1521))
```

Vi que o listener do host, no grid, estava configurado com hostname errado, corrija também:

```
[grid@gncasspl08329 admin]$ crsctl stop has -f
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'gncasspl08329'
CRS-2673: Attempting to stop 'ora.DATA_AUX.dg' on 'gncasspl08329'
CRS-2673: Attempting to stop 'ora.GRID.dg' on 'gncasspl08329'
CRS-2673: Attempting to stop 'ora.LISTENER.lsnr' on 'gncasspl08329'
CRS-2677: Stop of 'ora.DATA_AUX.dg' on 'gncasspl08329' succeeded
CRS-2677: Stop of 'ora.GRID.dg' on 'gncasspl08329' succeeded
CRS-2673: Attempting to stop 'ora.evmd' on 'gncasspl08329'
CRS-2673: Attempting to stop 'ora.ARCH.dg' on 'gncasspl08329'
CRS-2673: Attempting to stop 'ora.DATA.dg' on 'gncasspl08329'
CRS-2677: Stop of 'ora.ARCH.dg' on 'gncasspl08329' succeeded
CRS-2677: Stop of 'ora.DATA.dg' on 'gncasspl08329' succeeded
CRS-2673: Attempting to stop 'ora.asm' on 'gncasspl08329'
CRS-2677: Stop of 'ora.LISTENER.lsnr' on 'gncasspl08329' succeeded
CRS-2677: Stop of 'ora.evmd' on 'gncasspl08329' succeeded
CRS-2677: Stop of 'ora.asm' on 'gncasspl08329' succeeded
CRS-2673: Attempting to stop 'ora.cssd' on 'gncasspl08329'
CRS-2677: Stop of 'ora.cssd' on 'gncasspl08329' succeeded
CRS-2793: Shutdown of Oracle High Availability Services-managed resources on 'gncasspl08329' has completed
CRS-4133: Oracle High Availability Services has been stopped.

[grid@gncasspl08329 admin]$ cat listener.ora
#Backup file is /u01/app/grid/crsdata/gncasspl08329/output/listener.ora.bak.gncasspl08329.grid line added by Agent
# listener.ora Network Configuration File: /u01/app/19.19.0/grid/network/admin/listener.ora
# Generated by Oracle configuration tools.
```

```

LISTENER =
(DESCRIPTION_LIST =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = gncasspl08329.getnet.local)(PORT = 1521))
    (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
  )
)

```

```

ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER=ON          # line added by Agent
VALID_NODE_CHECKING_REGISTRATION_LISTENER=ON        # line added by Agent

```

```

[grid@gncasspl08329 ~]$ crsctl start has
CRS-4123: Oracle High Availability Services has been started.

```

```

[grid@gncasspl08329 ~]$ crsctl status res -t

```

Name	Target	State	Server	State details
Local Resources				
ora.ARCH.dg	ONLINE	ONLINE	gncasspl08329	STABLE
ora.DATA.dg	ONLINE	ONLINE	gncasspl08329	STABLE
ora.DATA_AUX.dg	ONLINE	ONLINE	gncasspl08329	STABLE
ora.GRID.dg	ONLINE	ONLINE	gncasspl08329	STABLE
ora.LISTENER.lsnr	ONLINE	ONLINE	gncasspl08329	STABLE
ora.asm	ONLINE	ONLINE	gncasspl08329	Started,STABLE
ora.ons	OFFLINE	OFFLINE	gncasspl08329	STABLE
Cluster Resources				
ora.biqa.db	1	ONLINE OFFLINE		Instance Shutdown,STABLE
ora.cssd	1	ONLINE ONLINE	gncasspl08329	STABLE
ora.diskmon	1	OFFLINE OFFLINE		STABLE
ora.evmd	1	ONLINE ONLINE	gncasspl08329	STABLE

Start do banco, já como Instance Name de BIQA:

```

[oracle@gncasspl08329 ~]$ vi db_profile
NOTA: alterei a linha:
export ORACLE_SID=BIQA

[oracle@gncasspl08329 dbs]$ pwd
/u01/app/oracle/product/12.1.0.2/dbhome_1/dbs
[oracle@gncasspl08329 dbs]$ cp cntrlBI.dbf cntrlBIQA.dbf

[oracle@gncasspl08329 ~]$ srvctl modify database -d BIQA -p /u01/app/oracle/product/12.1.0.2/dbhome_1/dbs/initBIQA.ora
[oracle@gncasspl08329 ~]$ srvctl start database -d BIQA

```

Adição do Service:

```

[oracle@gncasspl08329 ~]$ srvctl add service -d BIQA -s BIQASRV
[oracle@gncasspl08329 ~]$ srvctl start service -d BIQA -s BIQASRV
[oracle@gncasspl08329 ~]$ srvctl status service -d BIQA -s BIQASRV
Service BIQASRV is running
[oracle@gncasspl08329 ~]$

```

Criação do sqlnet.ora (por conta de senhas e pra deixar o wallet igual ao da Produção):

```
[oracle@gncasspl08329 ~]$ cd $ORACLE_HOME/network/admin
[oracle@gncasspl08329 admin]$ cat sqlnet.ora
SQLNET.EXPIRE_TIME = 2
SQLNET.INBOUND_CONNECT_TIMEOUT=120
SQLNET.ALLOWED_LOGON_VERSION=8
ENCRYPTION_WALLET_LOCATION =
  (SOURCE = (METHOD = FILE)
   (METHOD_DATA =
    (DIRECTORY = /u02/oracle/BI/wallet/)
   )
  )
```

Ultimo stop/start:

```
[oracle@gncasspl08329 admin]$ srvctl stop database -d BIQA
[oracle@gncasspl08329 admin]$ srvctl start database -d BIQA
```

```
SQL> alter system set wallet open identified by "xxxxx";
```

System altered.

```
SQL>
```

Evidência de conexão:

